

Effects of reforestation on near-surface saturated hydraulic conductivity in a managed forest landscape, southern Ontario, Canada

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ABSTRACT

Reforestation of previously deforested landscapes often increases soil organic matter content and porosity as the plantation forest ages, leading to increased soil saturated hydraulic conductivity (K_H). However, the time required for a reforested soil's K_H to recover to that of the original forest differs considerably between locations because of variations in forest, climatic and soil properties. We examined a chronosequence of sites on similar soils in a managed forest landscape in southern Ontario, Canada, ranging from open fields deforested in the late 19th century, through red pine (*Pinus resinosa*) stands of varying age, to 100+ year-old mixed conifer–hardwoods. Soil texture, organic matter content, bulk density, K_H , and overstory and understory forest characteristics were measured. For sites with similar soil textures, there was a general increase in K_H in the following sequence: open fields → red pine plantations → mixed conifer–hardwood stands. There were no significant changes in near-surface organic matter contents or soil bulk densities with forest age, and neither factor was related to temporal changes in K_H . Temporal trends in K_H were associated with increased understory density with forest age, suggesting that increases in K_H were driven by understory root development and greater macroporosity. It may take ~25 years before reforestation begins to affect soil infiltration characteristics at the stand scale and at least 40 years following planting before the K_H of soils under red pine plantations begins to equal that for undisturbed mixed conifer–hardwood stands. This recovery may be assisted by management activities such as thinning of red pine stands, which promotes understory development.

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KEY WORDS reforestation; saturated hydraulic conductivity; bulk density; organic matter content; forest properties; macroporosity; hydrologic recovery

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INTRODUCTION

Surface and near-surface soil saturated hydraulic conductivity (K_H) is a key hydrologic property that controls the partitioning of net precipitation into vertical (infiltration) and lateral (overland flow and throughflow) pathways (Ilstedt *et al.*, 2007; Zimmermann and Elsenbeer, 2009), with important implications for precipitation entry to and retention in subsurface storage as well as water fluxes to stream networks and the consequent production of stormflow and baseflow (Price *et al.*, 2010). A soil's K_H determines its steady-state infiltration capacity, which has been widely shown to be greater under vegetation cover compared with nonvegetated soils (e.g. Lee and Foster, 1991; Celik, 2005; Yimer *et al.*, 2008; Gonzalez-Sosa *et al.*, 2010; Price *et al.*, 2010). This difference has been attributed to the greater organic matter (OM) content, lower bulk density and higher macroporosity of vegetated versus nonvegetated soils because of macrofaunal activity and development of root networks in the former (Lee and Foster, 1991) combined with disturbance of previously vegetated soils that may reduce macroporosity in

the latter via compaction and removal of organic-rich upper soil layers through surface erosion (Price *et al.*, 2010).

Many workers (e.g. Li and Shao, 2006; Ilstedt *et al.*, 2007; Huang *et al.*, 2010; Mishra and Sharma, 2010) have concluded that the reintroduction of forest cover to previously deforested landscapes will result in an increase in soil K_H . The general expectation is that soils that have previously been degraded by deforestation will begin a process of hydrologic recovery (Hudson, 2000), moving through time towards the conditions found in natural forest soils. Previous observations of increases in K_H with forest age have been attributed to increased inputs of OM from litterfall, as well as greater macroporosity (Mapa, 1995; Deuchars *et al.*, 1999; Harden and Mathews, 2000; Godsey and Elsenbeer, 2002; Colloff *et al.*, 2010). However, studies based on a comparison of conditions in a small number of deforested, reforested and undisturbed forest stands do not allow the full definition of the temporal trajectory of K_H from disturbed to undisturbed conditions that is required in order to determine the time needed for complete hydrologic recovery of soil K_H . Conversely, those studies that have documented changes in soil hydraulic properties across a range of forest ages (e.g. Deuchars *et al.*, 1999; Zimmermann *et al.*, 2006; Hassler *et al.*, 2011) have often been conducted in tropical rain forests whose climatic, pedologic and ecologic properties may result in hydrologic recovery rates that differ greatly from those

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in other regions. Thus, there is an ongoing need for a detailed understanding of the impacts of reforestation on hydrological processes across a range of landscapes (Vose *et al.*, 2011). In particular, we need to understand the effects of reforestation on soil hydraulic properties, which may be as important as the changes in water use and soil moisture levels imposed by reforestation (Scott *et al.*, 2005).

A major challenge to acquiring this understanding is the lengthy period required for recovery of these hydraulic properties (McPherson and Timmer, 2002). Zimmermann *et al.* (2010) provided a rare example of the temporal response of K_H to conversion of forest to pasture and during secondary succession following pasture abandonment in the Brazilian Amazonia based on annual measurements at the same locations for several years. More typically, estimates of the hydrologic recovery of K_H have employed space-for-time substitution and the development of a chronosequence – a relationship wherein the soil-forming factors of climate, organisms, topography and initial materials are assumed to be constrained and only time changes (Schaetzl *et al.*, 1994). Construction of a chronosequence involves arranging soils of different age, and so from different locations, into a time sequence (Huggett, 1998). The approach assumes that the soils in the sequence represent successive stages of one or several pedogenic processes (such as the response to reforestation and forest growth) and that each soil passes through stages characterized by some preceding member of the successional sequence (Huggett, 1998). The assumption that the soils at each location had the same or similar initial properties is particularly problematic. These initial conditions may not be known; nevertheless, they can play a critical role in the recovery process (Scott *et al.*, 2005).

Presuming that a chronosequence can be derived, the resulting data can be used to estimate chronofunctions, which are mathematical relationships describing the rate of change of a soil property with time (Schaetzl *et al.*, 1994) and which have been used to quantify the recovery rates of disturbed soils (McPherson and Timmer, 2002). Chronofunctions have traditionally been estimated by linear regression using raw or transformed data (Schaetzl *et al.*, 1994). Such models may provide poor descriptors of a commonly observed response to environmental perturbation: a reaction time (the time required for the system to begin to respond to the perturbation) followed by the relaxation time during which the system completes its (often nonlinear) response to the perturbation (Huggett, 2011). Identification of the duration of both the reaction and relaxation times and the trajectory of system response during the latter are essential for understanding hydrologic recovery, and Schaetzl *et al.* (1994) emphasized the need to develop chronofunctions that are consistent with our current understanding of the pedologic system in question.

This study assesses whether the reforestation of a previously deforested and degraded landscape in south central Ontario has resulted in the hydrologic recovery of the soil's near-surface K_H . It seeks to meet two objectives:

1. to determine the temporal trajectory of K_H moving along a chronosequence from open fields through forest plantations of varying age to undisturbed mixed conifer–hardwood

stands that represent the native forest endpoint for this landscape; and

2. to assess the relative control of soil and vegetation properties on inter-site differences in K_H along this chronosequence.

STUDY AREA

The study was conducted in the Ganaraska Forest (GF; 44°5'N, 78°30'W; Figure 1), the largest block of continuous forest (44.5 km²) in southern Ontario. The GF is located on the crest and flanks of the Oak Ridges Moraine (ORM), an interlobate kame moraine that was deposited in patches from meltwater flowing in contact with moving or decaying glacier ice (Natural Resources Canada, 2002). The ORM consists of sand and gravel hills and high ridges comprising a complex interlaying of gravels, sand, silts, clays and minor diamictons (unsorted sand and/or coarser particles dispersed through a mud matrix – Allaby and Allaby, 2003) up to 150-m thick, underlain by a stony sandy silt to silty sand diamicton (Barnett *et al.*, 1998). Maximum elevation in the GF is 384 m a.s.l., 319 m above that of Lake Ontario to the south. The region has a humid mid-latitude climate (Köppen Dfb), and mean annual precipitation in the GF ranges from ~950 mm on its western edge to ~825 mm on its eastern edge (Ganaraska Region Conservation Authority, 2008), ~20% of which falls as snow. Annual evapotranspiration is ~530 mm (Brown *et al.*, 1968). There is no marked seasonality in precipitation (Buttle, 2011), and mean daily temperatures in the GF in January and July are –7.2 °C and 20.5 °C, respectively. The dominant soils in the GF are brunisolic grey brown luvisols (Soil Classification Working Group, 1998; Food and Agriculture Organization (FAO) equivalent: arenosol) belonging to either the Pontypool sand or Pontypool gravelly sand series, with sandy and sandy loam textures.

Much of the forest cover on the ORM was removed in the 19th century, and those areas converted to agriculture were used for growing crops such as hay and small grains (e.g. rye) or pasturing livestock (Richardson, 1944). Many of these farmed areas experienced severe wind and water erosion (Kelly, 1974) that removed the topsoil and exposed the calcareous parent material (Hill, 1976). Poor agricultural production led to the abandonment of many of these farms with fields left in fallow or unprotected, eroded conditions (Richardson, 1944; Howard *et al.*, 1996). Woodlot improvement, tree planting and erosion control were carried out on marginal and submarginal land in the GF beginning in 1948. Approximately 42% of the 44.5 km² of the GF was covered by mixed conifer–hardwood forest prior to reforestation [mainly red oak (*Quercus rubra* L.), with smaller amounts of white birch (*Betula papyrifera* Marsh.), sugar maple (*Acer saccharum* Marsh.) and other hardwoods, along with eastern white pine (*Pinus strobes* L.)]. Red pine (*Pinus resinosa* Ait.) was the main species planted with smaller plantings of white pine. Red and white pines were generally planted in combination at densities of ~2100 trees ha⁻¹. Reforestation was gradual between 1945 and 1950, the forest cover roughly doubled in the GF between 1950 and 1965, and there was minor infilling of

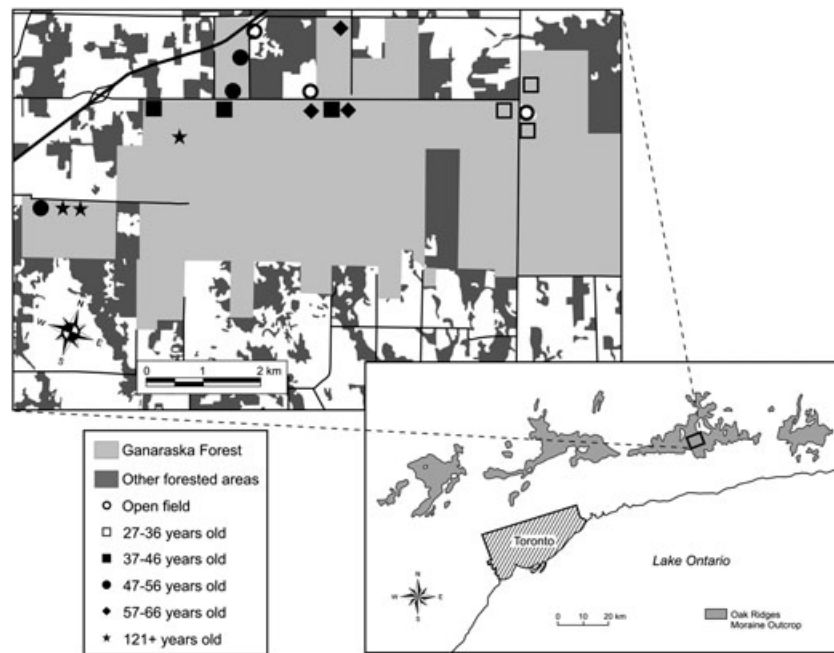


Figure 1. Location of study sites within and adjacent to the western and central portions of the Ganaraska Forest and the location of the Ganaraska Forest on the crest of the Oak Ridges Moraine, southern Ontario. Outcrops of the ORM are mapped according to Sharpe *et al.* (1999).

forest plantations after 1965. Reforestation essentially ended by 1985. Red pine plantations were generally thinned every 10–12 years, and the basal area (BA) at the time of thinning was approximately $28\text{--}32\text{ m}^2\text{ ha}^{-1}$. Approximately one third of the trees were removed during thinning, resulting in a reduction of BA to approximately $20\text{--}22\text{ m}^2\text{ ha}^{-1}$ (R. Penwell, Ganaraska Region Conservation Authority, Port Hope, personal communication, 2010). Trees were also pruned (live and dead branches removed from standing trees) up to 7.4 m from the ground in order to produce knot-free sawlogs. Red pine plantations serve as overstory for regenerating mixed hardwoods and conifers and are selectively thinned and harvested in order to convert the GF to a mature conifer–hardwood forest (Parker *et al.*, 2008).

METHODS

Site selection and sampling design

Site selection was based on the 1998 forest inventory for the GF and soil survey maps (Webber *et al.*, 1946). All sites were located on either Pontypool sand or Pontypool gravelly sand soils, and plantation sites had to consist of at least 50% red pine at time of planting. Areas in the GF reforested with red pine were generally either in fallow or had been extensively deflated by wind erosion prior to reforestation (McPherson and Timmer, 2002). The latter were readily identified from aerial photographs taken in early summer 1927 and were not included in this study. Reforested sites were grouped into decadal age categories (Table 1), and three sites were selected from each category. The initial condition of the reforested sites was abandoned fallow dominated by grasses (McPherson and Timmer, 2002), with the exception of one site (D10c, Table 1) that 1975 aerial photographs showed was in sparse mixed hardwood cover prior to reforestation. Three undisturbed

mixed hardwood–conifer forest sites with forest ages greater than 100 years were also selected, and open areas in and adjacent to the GF that had not been reforested were identified. Aerial photographs showed that three open sites had remained in agricultural production either as pasture (OPEN1 and OPEN3) or in small grains (OPEN2) as recently as 1975. At the time of the study, OPEN1 was managed as a tallgrass prairie site (Ganaraska Region Conservation Authority, Port Hope, personal communication, 2012) dominated by perennial ryegrass (*Lolium perenne* L.), common milkweed (*Asclepias syriaca* L.), goldenrod (*Solidago* L.) and poison ivy (*Toxicodendron radicans* L.). OPEN2 was in hay with a cover of alfalfa (*Medicago sativa*), timothy (*Phleum pratense*), smooth brome grass (*Bromus inermis* L) and red clover (*Trifolium pratense*), while OPEN3 was a lawn with a cover of fine fescues. These open sites were assumed to represent the initial conditions prior to reforestation (forest age = 0).

A level area was selected at each site, and a 12-m-radius circular sampling area was established around a central point. Eight random compass bearings and distances between 1 and 12 m were then used to define sampling locations. Saturated hydraulic conductivity was measured at the first five locations, while understory characteristics were measured at the other three (refer to the section on Forest cover measurements: overstory and understory). A pit was dug at the central point to determine the site's soil properties. Thus, there were 18 sites (Table 1) with five measurements of K_H , three measurements of understory characteristics and one soil pit at each site.

Saturated hydraulic conductivity

Measurement of K_H was made using a double ring cylinder infiltrometer (Hills, 1970), with an inner ring diameter of 20 cm and an outer ring diameter of 30 cm. After the depth

Table I. Forest age, mean K_H and selected soil and vegetation properties for the study sites.

Site	Forest age (years)	Mean K_H (mm h ⁻¹)	% OM at 0.05-m depth	ρ_b at 0.05-m depth (kg m ⁻³)	Maximum ρ_b in profile (kg m ⁻³)	Mean % gravel	Mean % sand	Mean % silt + clay	Mean tree height (SD) (m)	Basal area (m ² ha ⁻¹)	Number of saplings (ha ⁻¹)	Mean % understory (SD)
<i>OPEN1</i>	0	223	8.1	824	1666	0.2	68.2	31.6	0	0	0	0
<i>OPEN2</i>	0	180	3.7	1331	1721	1.3	32.9	65.8	0	0	0	0
<i>OPEN3</i>	0	792	4.0	1193	1694	0.6	59.0	40.4	0	0	0	0
<i>152j</i>	27	2952	3.4	1300	1798	3.8	65.5	30.7	11 (0.7)	33	420	0.7 (1.2)
<i>152e</i>	27	266	2.4	1293	1858	1.4	62.2	36.5	11 (1.9)	31	0	0
<i>45b</i>	30	792	2.3	1382	1719	0.3	90.9	8.8	11 (0.3)	39	22	0
<i>D10c</i>	39	6768	10.0	836	1716	7.8	76.9	15.4	12 (3.1)	31	928	69.3 (17.7)
<i>9b</i>	42	590	3.7	1165	1591	0.1	57.5	42.4	15 (5.6)	21	1857	39.7 (20.0)
<i>D9a</i>	43	1980	3.5	1361	1682	0.3	85.3	14.4	15 (3.4)	22	1216	46.0 (21.2)
<i>37b</i>	50	4824	3.2	1537	2001	28.0	63.6	8.4	11 (5.9)	46	1415	23.0 (18.2)
<i>3f</i>	53	2808	3.8	1354	1818	8.6	73.9	17.5	14 (6.0)	25	1105	26.7 (14.6)
<i>7e</i>	54	1800	3.2	1171	1670	0.3	83.2	16.5	14 (5.8)	15	1614	54.7 (26.4)
<i>43b</i>	61	2488	1.4	1571	1873	1.1	74.0	24.9	17 (4.2)	39	729	23.0 (20.7)
<i>8b</i>	61	828	2.1	1404	1643	0.3	75.0	24.7	11 (5.4)	35	1547	26.7 (13.1)
<i>2a</i>	64	810	3.1	1289	1652	0.1	70.8	29.1	14 (5.8)	32	1194	35.7 (6.7)
<i>22b</i>	124	4104	3.9	1139	1734	0.9	79.3	19.8	13 (4.8)	31	818	65.7 (17.9)
<i>36c</i>	139	3096	3.5	1126	1693	2.0	72.8	25.2	9 (3.6)	26	1967	28.7 (31.0)
<i>37c</i>	149	1008	4.7	1259	1733	1.2	66.0	32.8	12 (6.7)	25	1481	41.7 (14.6)

Sites considered to have similar soil textural properties (refer to text) are indicated in italics. Soil texture %s may not sum to 100 because of rounding.

of the litter and organic layers overlying the mineral soil was measured, these were removed, and the rings were inserted to a depth of 5 cm. A constant head device maintained a 5-cm depth of water in the inner ring during the infiltrometer tests, while water was supplied manually to the outer ring to maintain ponded conditions during the test. Plots of cumulative infiltration versus time since the start of the test were constructed, and K_H was taken as the slope of the linear portion of the cumulative infiltration curve (Buttle and House, 1997). Water inputs during the tests suggest that field saturation (incomplete saturation due to air entrapment – Reynolds, 1993) of at least the upper 0.75–1.0 m of the soil profile occurred, such that the derived K_H value represents the conductivity of the least permeable portion of this depth interval.

Soil properties

Horizontal soil cores beginning at 0.05-m depth were taken at 0.1-m intervals from the soil pit to a depth of 0.75 m. Bulk density (ρ_b) was determined for each core and used to estimate porosity (η) assuming a particle density of 2650 kg m^{-3} . Soil OM content was determined by loss on ignition (Tiessen and Moir, 1993). Observations were made of the root depth and diameter of the grasses at the OPEN sites and understory vegetation at the forest sites. Gravel content (particles >2-mm diameter) of each core sample was determined by sieving, and the sand, silt and clay portions of the fraction ≤ 2 -mm diameter were determined using a Horiba LA-950 V2 Laser Scattering Particle Size Distribution Analyzer.

Forest cover measurements: overstory and understory

Height, species and BA of all trees greater than 2.5-cm diameter at breast height (dbh) within the 12-m-radius sampling area at each site were recorded. Overstory density was calculated as the sum of the BAs of all trees within the sampling area divided by the sampling area (Shiver and Borders, 1996). The number of trees within the sampling area with dbh <2.5 cm and heights ≥ 2.5 m was determined to obtain the density of the younger overstory component. Understory species composition, height and density at each site were determined by three randomly located 1-m² quadrat counts within the sampling area (Mueller-Dombois and Ellenberg, 2003).

Statistical analyses

Grubbs' (1969) test for outliers was applied to K_H measurements to identify outliers, and the Shapiro–Wilk test was used to test all variables for normality. In an attempt to control for inter-site differences in soil texture, cluster analysis was used to group sites with similar soil textural properties. Properties examined were average % gravel content, average % silt+clay content, and the maximum range in % silt+clay content with depth in the profile. Soil K_H has been found to increase with mean % gravel content (Ravina and Magier, 1984, Sauer and Logsdon, 2002), which has been attributed to larger pore sizes associated with increased gravel content in the profile.

Similarly, greater % silt+clay content in the profile has been correlated to a decrease in average pore diameter and smaller K_H (Norris and Fidler, 1965; Masch and Denny, 1966). The maximum range in % silt+clay content with depth was used to distinguish those sites with relatively little change in the vertical profile of % silt+clay content from those that might have an impeding layer in the profile that would control the K_H determined from the ring infiltrometer measurements. Cluster analysis was conducted using a single linkage amalgamation rule and squared Euclidean distances as the distance measure (StatSoft, 2000). Because many variables were non-normally distributed, the nonparametric Spearman rank correlation coefficient (r_s) was used to examine relationships between variables for the complete data set and for the subset based on similar soil textural properties. The nonparametric Mann–Kendall (Z) test (McCuen, 2003) was also used to assess monotonic temporal trends in K_H , soil and vegetation properties. A chronofunction relating normalized mean K_H at each site to forest age was estimated by linear regression. All statistically significant relationships are reported at the $p=0.05$ level.

Hydrologic recovery

We followed Hudson's (2000) method of estimating hydrologic recovery (HR) for K_H :

$$HR(\%) = 100 - \left[\left(\frac{K_{Hx} - K_{Hu}}{K_{Ho} - K_{Hu}} \right) \times 100 \right] \quad (1)$$

where K_{Hx} is the mean K_H for a given site, and K_{Hu} and K_{Ho} are the K_H values for undisturbed forest stands and open sites that had not been reforested, respectively. K_{Hu} was assumed to be represented by the average K_H for the three 100+ year-old mixed conifer–hardwood stands, which had an average age of 137 years. This approach assumes a linear change in HR index between the recovery indices for 100% (undisturbed) and 0% (unforested) recovery states (Hudson, 2000). A Chapman–Richards curve of the form

$$HR = a \left[1 - e^{-b(x-d)} \right]^c \quad (2)$$

was found to provide a good fit to HR results for rainfall interception and snowmelt in coastal BC (Hudson and Horel, 2007). This curve was used to fit the K_H HR values, where x is forest age (open sites are assumed to have age = 0 year), a is the asymptote for recovery (assumed to be 100%), b and c are curve fitting parameters and d is the threshold age where recovery of K_H begins. Coefficients in Equation (2) were estimated by optimizing the fit between observed and predicted HR values in terms of the Nash–Sutcliffe efficiency (Buttle *et al.*, 2005).

RESULTS

Saturated hydraulic conductivity

The Grubbs test revealed one outlier at each of sites 2a and 9b; these were removed prior to subsequent analyses. Mean K_H varied from 180 to 6768 mm h^{-1} across all sites

(Figure 2, Table 1). Maximum K_H values were similar to those estimated by Buttle and House (1997) for sandy forest soils from ring infiltrometer measurements in the field and constant head permeameter measurements of soil cores. Of the three smallest mean K_H values, two were from open fields representing the 'pre-plantation' condition (OPEN1 and OPEN2), while the third was from the youngest plantation age category (152e). In contrast, mean K_H values at other sites exceeded 4000 mm h^{-1} and were spread across a wide range of forest age categories. Two pine plantation sites (D10c and 37b) had mean K_H values in excess of those found in the undisturbed mixed conifer–hardwood stands. The former was under mixed hardwood forest cover prior to reforestation. Variability in site K_H tended to be least in the undisturbed sites (mean coefficient of variation $CV = 0.19$; range = 0.16 – 0.24), was slightly greater in the open fields (mean $CV = 0.37$; range = 0.31 – 0.50) and was greatest in the red pine plantations (mean $CV = 0.46$; range = 0.15 – 0.94).

Soil texture

Most soils were generally sandy, and only one site (OPEN2) had a mean sand content less than 50% (Table 1). Clay and gravel contents were generally small across all sites, with the exception of mean gravel contents $>3\%$ at four sites (152j, C10c, 37b and 3f). While many profiles showed little change in texture with depth, others showed distinct gravel and clay lenses (37b), significant silt lenses (43b) or marked increases in silt content with depth (OPEN2, OPEN3 and 9b). Cluster analysis indicated that 13 sites had % gravel, % silt + clay and maximum range in % silt + clay values that distinguished them from the remaining five sites (OPEN2, OPEN3, 9b, 37b and 43b) at a linkage distance of 160. Nevertheless, three of these sites (3f, D10c and 152j) had % gravel values in excess of the remaining 10 sites (mean gravel content = 0.7%). Their removal resulted in a subset of 10 sites (henceforth referred to as the sand-dominated subset) considered to have similar soil textural properties (Table 1): a gravel fraction $<2\%$, a mean silt + clay fraction between 9% and 33% and a maximum range in silt + clay fraction with depth in the profile between 9% and 33% .

Organic matter content, root characteristics and bulk density

All sites showed a general decrease in soil OM content with depth, with little inter-site difference in OM contents below depths of 0.2 m . Near-surface OM contents (represented by samples at 0.05-m depth) varied considerably about the mean value of 3.8% (Table 1). The large near-surface OM content at D10c may reflect its mixed hardwood forest cover prior to reforestation. Roots of the tallgrass prairie species at OPEN1 extended up to 0.3-m depth and consisted mainly of vertically oriented thin ($<1\text{-mm}$ diameter) fibrous roots and larger taproots (5- to 10-mm diameter, milkweed). Vertically oriented thin ($<1\text{-mm}$ diameter) fibrous roots extending to 0.05- to 0.07-m depths were found at OPEN2 and OPEN3. More extensive root systems at the OPEN1 site may account its larger near-surface OM contents relative to the other OPEN sites. In addition, the OPEN1 site had been burned annually for several years as part of its management as a tallgrass prairie site (Ganaraska Region Conservation Authority, Port Hope, personal communication, 2012). Burning has typically been found to decrease surface OM contents via combustion (Neff *et al.*, 2005); however, it may also alter decomposition rates in such a way as to promote OM accumulation in near-surface soil. Woody understory root systems in the forest sites consisted of central taproots (10- to 15-mm diameter) extending to 0.15- to 0.20-m depth, coupled with shallow (0.05- to 0.07-m depth) horizontal roots (1- to 3-mm diameter). Few tree roots were observed at 0.7-m depth at the forest sites. Bulk density at all sites exhibited expected increases with depth, with relatively little change beyond depths of 0.25 m . Much of the inter-site variation in near-surface ρ_b reflected the strong inverse correlation with OM content in near-surface soil (Tables 1 and 2).

Vegetation characteristics

Mean tree height varied considerably between forested sites with no obvious relationship between height and forest age (Table 1). Nevertheless, the relative variability in tree height increased with forest age. BAs were typically between 20 and $35 \text{ m}^2 \text{ ha}^{-1}$, with all but one site exceeding $20 \text{ m}^2 \text{ ha}^{-1}$ (Table 1). BAs for younger red pine plantations

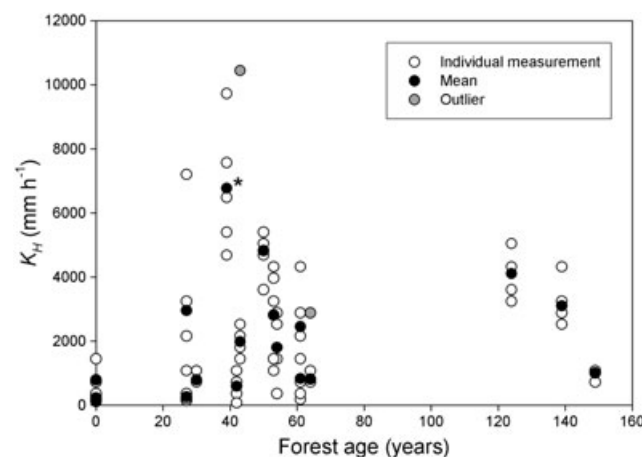


Figure 2. Saturated hydraulic conductivity (K_H) measured at each site versus forest age. The mean K_H at each site is indicated. Outlier K_H measurements were not used in calculating the site mean K_H . The mean K_H indicated by * had a mixed hardwood forest cover prior to reforestation, while all other reforested sites (forest ages from 27 to 64 years) were abandoned fallow fields prior to reforestation. Refer to text for details.

Table II. Spearman rank correlation coefficient matrix for selected variables from the sand-dominated site subset.

Variable	Forest age	Mean K_H	$OM_{0.05}$	$\rho_{b0.05}$	ρ_{bMAX}	BA	Mean % cover	No. of saplings	Mean % gravel	Mean % sand	Mean % silt + clay
Forest age	—										
Mean K_H	0.70	—									
$OM_{0.05}$	0.23	0.25	—								
$\rho_{b0.05}$	-0.18	-0.20	-0.83	—							
ρ_{bMAX}	0.09	0.18	0.20	0.94	—						
BA	0.09	-0.16	-0.77	0.67	0.05	—					
Mean % cover	0.58	0.83	0.33	-0.21	0.07	-0.34	—				
No. of saplings	0.67	0.66	-0.04	-0.03	-0.32	-0.13	0.49	—			
Mean % gravel	0.39	0.50	0.21	-0.19	0.70	-0.14	0.20	0.30	—		
Mean % sand	-0.08	0.41	-0.30	0.32	-0.16	0.14	0.33	0.24	-0.25	—	
Mean % silt + clay	0.08	-0.41	0.30	-0.32	0.16	-0.14	-0.33	-0.24	0.25	-1.00	—

Statistically significant values are italicized.

$OM_{0.05}$, organic matter content at 0.05 m depth; $\rho_{b0.05}$, bulk density at 0.05-m depth; ρ_{bMAX} , maximum bulk density in soil profile; BA, basal area.

(e.g. 45b and 152j) exceeded those for some older sites (e.g. 7e and 9b), which reflects thinning of the latter. This, combined with $BA \leq 30 \text{ m}^2 \text{ ha}^{-1}$ for undisturbed mixed conifer–hardwood stands, resulted in no statistically significant increase in BA with forest age. In addition, unlike older red pine plantations, plantations in the youngest age category (152j, 152e and 45b) had not been thinned or pruned. Visual observations suggested that this resulted in greater canopy closure and less light transmission to the forest floor, which corresponds with the small sapling numbers and mean % understory cover at these sites (Table 1). Saplings at all forest sites consisted of a mixture of sugar maple, hop hornbeam (*Ostrya virginiana*) and trembling aspen (*Populus tremuloides*). Major understory species in the red pine plantations were dogwood (*Cornus alternifolia*), raspberry (*Rubus idaeus* L.), hop hornbeam and trembling aspen. In addition to hop hornbeam, sugar maple and trembling aspen, understory cover in the undisturbed mixed conifer–hardwood stands also contained white ash (*Fraxinus americana*). There was considerable variability in sapling number and mean % understory cover between older pine plantation sites and the undisturbed mixed conifer–hardwood stands; nevertheless, both variables showed statistically significant increases with forest age across all sites ($r_s = 0.72$ and 0.65 , respectively). These trends were driven partly by the 0 values for sapling number and mean % understory cover for the open field sites, which all had uniform grass cover.

Inter-site variations in K_H

There was a statistically significant increase in mean K_H ($r_s = 0.50$), sapling number ($r_s = 0.72$) and mean % understory cover ($r_s = 0.65$) with forest age when all sites were considered. BA ($r_s = 0.49$), mean % understory cover ($r_s = 0.58$), mean % gravel content ($r_s = 0.60$) and mean % silt + clay content ($r_s = -0.65$) were significantly correlated with mean K_H across all sites. The largest mean K_H values in the pine plantations were associated with sites with the greatest mean % gravel contents (Table 1, Figure 3a), and although mean % gravel content was correlated with

maximum ρ_b in the soil profile ($r_s = 0.76$), there were no significant relationships between mean K_H and either near-surface or maximum ρ_b . Mean K_H also showed a statistically significant increase with forest age (Table 2, Mann–Kendall $Z = 2.06$) for the sand-dominated subset of sites, and the chronofunction relating normalized (cube-root) mean K_H to forest age for these sites (Figure 3a) was significant at the $p = 0.05$ level. The only other variable significantly correlated with forest age for the sand-dominated sites was sapling number (Mann–Kendall $Z = 1.99$). Sapling number and mean % understory cover were the only properties significantly correlated with mean K_H for these sites (Table 2). Mean K_H was not significantly correlated with any soil textural properties for the sand-dominated sites.

Hydrologic recovery

Figure 3b shows observed and predicted HR values for the sand-dominated sites. The optimized form of Equation (2) has a Nash–Sutcliffe efficiency value of 0.46, which indicates an acceptable fit to the data (Moriassi *et al.*, 2007). Unlike a linear regression of HR on forest age, which would imply a progressive increase in K_H immediately following the initiation of reforestation, both the HR values and the optimized Chapman–Richards curve suggest a threshold forest age (26 years) that must be exceeded before recovery of K_H occurs.

DISCUSSION

Changes in K_H with time

Mean K_H for across all sites and for sites with similar soil textures showed a significant increase with forest age from open grassed sites to undisturbed mixed conifer–hardwood stands. This is consistent with the results from other reforestation scenarios that have been based on chronosequences obtained by using time-for-space substitution. Colloff *et al.* (2010) reported that reforestation of sandy loam Spodosols in Australia with eucalyptus and acacia trees led to a doubling of infiltration rates after

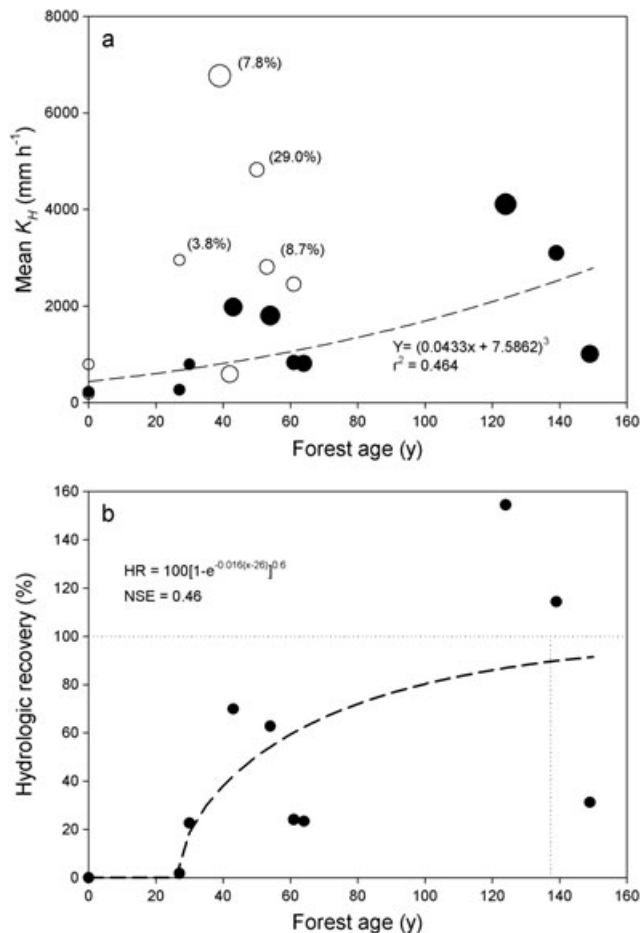


Figure 3. (a) Mean saturated hydraulic conductivity K_H versus forest age. Solid circles and the line of best fit are for the sand-dominated sites. Symbol size is proportional to mean % understory cover at the site: the smallest symbol represents a mean understory cover of 0%, while the largest symbol represents a mean understory cover of 69%. Values in parentheses indicate the mean % gravel content at that site. (b) Hydrologic recovery versus forest age. The horizontal dotted line indicates 100% hydrologic recovery, while the vertical dotted line indicates the mean age (137 years) of the undisturbed mixed conifer-hardwood stands.

11 years: $\sim 35 \text{ mm h}^{-1}$ for previously deforested paddock sites to $\sim 70 \text{ mm h}^{-1}$ for 11- to 20-year-old plantations. Deuchars *et al.* (1999) found that rainforest regeneration on volcanic Andisols in Costa Rica resulted in an increase in K_H from $\sim 3.6 \text{ mm h}^{-1}$ in deforested pastures to $\sim 360 \text{ mm h}^{-1}$ for 15-year-old plantations and $\sim 3600 \text{ mm h}^{-1}$ for 40-year-old plantations and primary rainforest. Both studies suggest that 30–40 years may be required for infiltration capacities to return to those of undisturbed forest conditions, and the present results suggest that it may take in the order of 40 years before the mean K_H of pine plantations in the GF begins to equal or exceed the lowest values found in undisturbed mixed conifer-hardwood stands. In contrast, Harden and Mathews' (2000) examination of the effects of reforestation on the hydraulic properties of Ultisols near an abandoned copper mine and smelter in Tennessee found that replanting with Loblolly pine (*Pinus taeda*) led to increased infiltration rates but that upwards of 70 years were needed for the soils to exhibit infiltration characteristics similar to those of the original mixed hardwood stands. Emissions of SO_2 from the

smelter that continued until the late 1980s caused acidification and vegetation mortality in the study area, which may have slowed the rate of forest regrowth and delayed hydrologic recovery (Harden and Mathews, 2000).

The chronofunction relating mean K_H to forest age (Figure 3a) implies that reforestation may result in an almost-immediate improvement of infiltration characteristics in the GF. However, 1965 aerial photographs of young (< 10 years) red pine plantations show open areas extending for several metres between the tree rows in which the soil's hydraulic properties are unlikely to have been affected by canopy contributions of OM or tree root extension. A rapid response of K_H to reforestation is also inconsistent with the time lag between the initiation of forest recovery and the response of soil hydraulic properties suggested by both the temporal trajectory of HR values in Figure 3b and previous studies of soil response to reforestation. Thus, Wilde (1964) noted that red pine plantations younger than 20 years did not greatly affect soil characteristics; however, prolonged litter deposition and nutrient leaching from the canopy of older plantations resulted in pronounced soil amelioration. Zimmermann *et al.* (2006) found that 15 years of secondary succession in a tropical montane forest region sufficed to restore belowground K_H to that of old-growth forest but was insufficient to result in the recovery of surface infiltrability. Zimmermann and Elsenbeer (2008) noted a very slight recovery of K_H at 12.5- and 20-cm depths at the same location after 10 years of natural succession on an abandoned pasture. Conversely, K_H for a nearby 25-year-old pine plantation on abandoned pasture had experienced full hydrologic recovery. De Moraes *et al.* (2006) measured K_H on abandoned pasture in Amazonia following 12 years of natural succession that was undistinguishable from that on active pasture, while Zimmermann *et al.* (2010) noted only a slight but statistically nonsignificant increase in surface infiltrability and K_H at 12.5- and 20-cm depths in an abandoned pasture in the Brazilian Amazonia that had undergone 7 years of secondary succession. Zimmermann *et al.* (2010) characterized the changes in K_H as a result of forest conversion to pasture followed by forest regeneration on abandoned pasture as an asymmetric response to disturbance and recovery: a rapid reduction in K_H due to pasture initiation followed by a disproportionately longer time required for K_H to return to pre-disturbance conditions. Our observations of K_H response to reforestation in the GF are consistent with this conceptual model.

There was considerable scatter about the chronofunction and HR curves (Figure 3a,b), which may be attributed in part to the degree to which the chronosequence satisfies the assumptions underlying the space-for-time substitution used in its derivation. The GF sites have similar climate, topography, surficial geology and soil types. Although there were marked variations in soil texture between sites, we attempted to account for these by identifying a sand-dominated subset with similar textures. However, persisting variations in soil properties within this subset were likely responsible for some of the inter-site differences in mean K_H . A more difficult issue to resolve is the extent to which the initial conditions of the reforested sites were similar. We assumed that sites were abandoned fallow fields immediately prior to reforestation with hydraulic properties

similar to present day proximal agricultural areas and thus no significant change in a site's infiltration characteristics during the transition from agricultural use to abandoned fallow land. This is supported by the observation of Zimmermann *et al.* (2010) of a rapid decrease in near-surface K_H with an initial conversion from forest to pasture in the Brazilian Amazonia followed by an insignificant change in surface infiltrability during the first 7 years of abandonment. Nevertheless, some of the scatter in the hydrologic recovery of K_H with forest age in the GF may reflect variations in such initial conditions as the type and intensity of agricultural activity between regenerating sites. Information on these initial conditions is not available; nevertheless, such conditions can partly govern the temporal trajectory of K_H recovery (Zimmermann *et al.*, 2010).

Controls on the hydrologic recovery of K_H in the GF

Grouping of sites based on similar soil textural properties allowed us to focus on other controls on K_H that may also change as the landscape makes the transition from grassed fields to mature mixed conifer–hardwood forest. Previous work on the recovery of K_H with forest regeneration has focused on the role of decreases in soil ρ_b associated with OM accumulation from greater litterfall, either in isolation or in combination with increased soil macroporosity due to root development and macrofaunal activity. Results from the GF suggest that reforestation has not resulted in significant changes in near-surface OM content and ρ_b . This contrasts with a significant decrease in surface ρ_b with forest age for a nearby location on the ORM reforested with red pine, which was attributed to soil OM accretion and increased root and faunal activity (McPherson and Timmer, 2002). Our results are also inconsistent with the increased OM contents found by Harden and Mathews (2000) from ~5% in pre-plantation soils to ~16% in undisturbed forest sites in Tennessee and a general increase in OM contents of near-surface soils observed by Deuchars *et al.* (1999) for the transition from 15- to 20-year-old plantations and then to primary rainforest. However, the absence of significant correlations between mean K_H and either near-surface OM content or ρ_b suggests that factors that promote the development of soil macroporosity may exert a relatively greater control on temporal changes in K_H .

Vegetation properties and soil macroporosity

Soil macroporosity has a major influence on a soil's infiltration characteristics (Beven and Germann, 1982), and greater macroporosity can increase the ability of the soil profile to conduct flow (e.g. Watson and Luxmoore, 1986; Smettem, 1987; Wilson and Luxmoore, 1988; Buttle and McDonald, 2000). These macropores can form through the development of root systems, which tend to produce larger, more continuous and less tortuous conduits for water than the smaller pores that result from soil aggregation promoted in part by the incorporation of litterfall into the soil's near-surface OM content (Brady and Weil, 2010). Both saplings and understory vegetation can develop extensive root networks in the upper soil, and sapling numbers and mean % understory cover were found to increase with forest age in the GF and elsewhere on the

ORM (e.g. McPherson and Timmer, 2002). However, only the latter variable showed a significant correlation with mean K_H . Similar controls of understory cover on K_H have been observed in other reforestation and afforestation studies. Deuchars *et al.* (1999) found that increased K_H following rainforest regeneration was accompanied by substantial (up to 20%) increases in the porosity of surface soils, which they attributed to understory root development. Both Sorriso-Valvo *et al.* (1995) and Navas *et al.* (2008) found a strong correlation between the spatial distribution of K_H and understory characteristics. The significant control of understory development on K_H was highlighted by Sorriso-Valvo *et al.* (1995), who observed much lower infiltration rates on sites afforested with 6-year-old eucalyptus trees compared with sites with no forest cover. They attributed this counter-intuitive result to the inhibition of understory growth by eucalyptus, whereas nonforest sites were covered by dense grasses and shrubs that could promote macroporosity via root growth.

Forest management and K_H

Although our results suggest that the increased mean K_H with forest age may be attributed in part to the development of understory cover and its potential role in increasing soil macroporosity, they also indicate that this relationship depends in part on how the red pine plantations in the GF are managed. As noted earlier, the youngest plantations (which had few or no saplings and minimal understory cover) had not been thinned or pruned at the time of the study, leading to reduced light levels at the forest floor, which may have inhibited seedling germination (Ehrenfeld, 1980; Moore and Vankat, 1986). Parker *et al.* (2008) showed that thinning of nearby red pine plantations had a significant positive effect on the survival and growth of understory vegetation, which they attributed to increases in light, moisture and nutrient availability, and mean air and soil temperature. Visual observations indicated that the younger, unthinned stands had thick litter layers of pine needles relative to thinned stands, and Parker *et al.* (2008) noted that red pine plantations with thick litter layers also require some form of forest soil disturbance to create seedbeds and stimulate the soil seed bank. Greater sapling numbers and understory cover in older red pine plantations in the GF may reflect soil scarification that occurred during thinning operations, which could initiate understory development by bringing seeds to the soil surface and exposing them to the environmental conditions necessary for germination (Luken, 1990). Thus, in addition to the production of timber and the promotion of a vigorous understory of mid-tolerant tree species that encourages the transition to a mature mixed conifer–hardwood forest (Parker *et al.*, 2008), thinning of red plantations in the GF may also contribute to an increase in soil K_H .

Hydrological implications of reforestation in the GF

Several studies have assessed the hydrologic significance of changes in K_H associated with differences in land use (e.g. Price *et al.*, 2010) or during forest regeneration on

previously disturbed sites (e.g. Zimmermann *et al.*, 2006; Zimmermann and Elsenbeer, 2009; Bonell *et al.*, 2010) by comparing K_H values with rainfall intensity–duration–frequency (IDF) information. Such comparisons indicate the potential for storms with rainfall intensities in excess of soil K_H to generate overland flow and near-surface lateral flow through the soil above an impeding layer in the profile. The IDF curves from the city of Peterborough (~40 km northeast of the GF) estimate the 100-year return period maximum 0.5- and 1-h rainfall intensities for the region to be in the order of 80 and 50 mm h⁻¹, respectively (Buttle and Lafleur, 2007). These values are well below the smallest mean K_H observed at any of the GF sites (180 mm h⁻¹ at OPEN2), suggesting that surface and near-surface lateral flow in this landscape is unlikely at present. However, this flow was more prevalent along the ORM prior to reforestation, and the hydrologic effects of the establishment and growth of red pine plantations on the infiltration properties of soils in the GF partly explain changes in streamflow characteristics observed in the Ganaraska River basin, which drains much of the GF (Buttle, 2011). Reforestation was associated with decreases in some peak streamflow metrics and increases in annual minimum daily runoff. Peak streamflows in the Ganaraska River basin are associated with rain-on-snow events during spring snowmelt, and increased K_H due to reforestation would decrease peak-flows by promoting infiltration of a greater fraction of water inputs to the soil surface, thus reducing overland flow to stream channels. Conversely, increased infiltration of water inputs during snowmelt would augment soil water and groundwater storage, which could then contribute to enhanced baseflows during the subsequent summer period.

CONCLUSIONS

Our results indicate that reforestation of a previously degraded landscape in southern Ontario has resulted in the hydrologic recovery of the soil's saturated hydraulic conductivity. However, they also suggest that inducing this hydrologic recovery is not simply a matter of putting trees back on the landscape. Thus, the mean K_H for a 27-year-old unmanaged red pine plantation was essentially the same as that of an open field with similar soil texture that had been under continuous grass cover for more than 100 years. Furthermore, it may take 40 years or more for the K_H of red pine plantations to equal that of undisturbed mixed conifer–hardwood stands. In addition, increases in K_H in forest sites were correlated with greater forest understory development, likely via the associated increase in soil macroporosity. Initiation and growth of understory cover can be promoted by such forest management activities as stand thinning and pruning. The hydrologic recovery of soil K_H following reforestation appears to have implications for the runoff regime of streams in this landscape, and these should be considered when assessing the impacts of changes in both climate and land use on this region's water resources.

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